

NEW HORIZON COLLEGE OF ENGINEERING

Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC
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DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

“AI-Powered Internship Recommendation and Verification System using SBERT and Blockchain”

Major Project [22ISE74]

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Introduction

- Internship recommendation helps students identify opportunities aligned with their skills, interests, and career goals.
- Traditional recommendation systems mainly rely on keyword matching, often failing to capture the contextual meaning of resumes and internship descriptions.
- This project proposes an AI-Powered Internship Recommendation and Verification System using Sentence-BERT (SBERT) and Cosine Similarity for semantic matching.
- The platform supports cross-border internship discovery using country, location, work mode, duration, stipend, and eligibility information.
- It incorporates suspicious internship detection, company email verification, and blockchain-based credential verification.
- By combining AI, NLP, Semantic Matching, and Blockchain Technology, the system aims to provide accurate, trustworthy, and efficient internship recommendations.

Problem Definition

- Students often struggle to find internships that match their skills, qualifications, and career interests.
- Existing recommendation systems rely heavily on keyword-based matching and frequently generate irrelevant recommendations.
- Cross-border internships introduce additional challenges related to eligibility, location preferences, and visa requirements.
- The growing number of suspicious internship postings creates trust and security concerns for students.
- Most current platforms lack proper company verification and secure credential validation mechanisms.
- Therefore, an intelligent system is required to provide accurate recommendations, detect suspicious postings, and ensure secure verification of internship credentials.

Literature Survey

Sl. No	Author / System	Methodology / Focus	Project Relevance / Role
1	Reimers & Gurevych (2019) - Sentence-BERT	Sentence embeddings using Siamese BERT networks.	Basis for representing resumes and internship descriptions in a shared semantic space.
2	Devlin et al. (2019) - BERT	Bidirectional Transformer pre-training for language understanding.	Foundational transformer approach behind the NLP/ embedding layer.
3	Reimers & Gurevych (2020) - Multilingual Sentence Embeddings	Knowledge distillation approach for multilingual sentence embeddings.	Supports the cross-border / multilingual direction of semantic matching.
4	Nousias et al. (2022) - Blockchain Verification	Process-aware blockchain-based verification of academic qualifications.	Informs the credential verification and on-chain record design.

Literature Survey

Sl. No	Author / System	Methodology / Focus	Project Relevance / Role
5	Rageshnithin et al. (2025) - Cross-Border Credential Verification	ZKP and blockchain perspectives for global credential verification.	Supports the project direction for secure, verifiable credentials.
6	Alonso et al. (2025) - Job Matching & Skill Recommendation	Transformers and O*NET-oriented matching and skill recommendation.	Provides a research basis for semantic job/skill matching.
7	Poncio (2024) - Internship Profile Matching Review	Systematic review of techniques for job recommender and internship profile matching.	Provides broader context for internship recommendation design.
8	El-Deeb et al. (2025) - E-Recruitment Recommendations	Text summarization techniques for e-recruitment recommendations.	Relevant to text-processing and recommendation workflows.
9	El-Deeb et al. (2025) - Explainable & Scalable Job Recommendation	Transformer-based summarization, cross-encoder similarity, and ensemble learning.	Highlights explainability and scalability considerations.
10	Taneja et al. (2025) - Fraud-BERT	Transformer-based, context-aware online recruitment fraud detection.	Research basis for the suspicious-posting risk module.
11	Srilakshmi et al. (2026) - Recruitment Fraud Detection	BERT, RoBERTa, SMOBD, and 2D CNN based fraud-detection framework.	Provides recent fraud-detection context for future classifier evaluation.

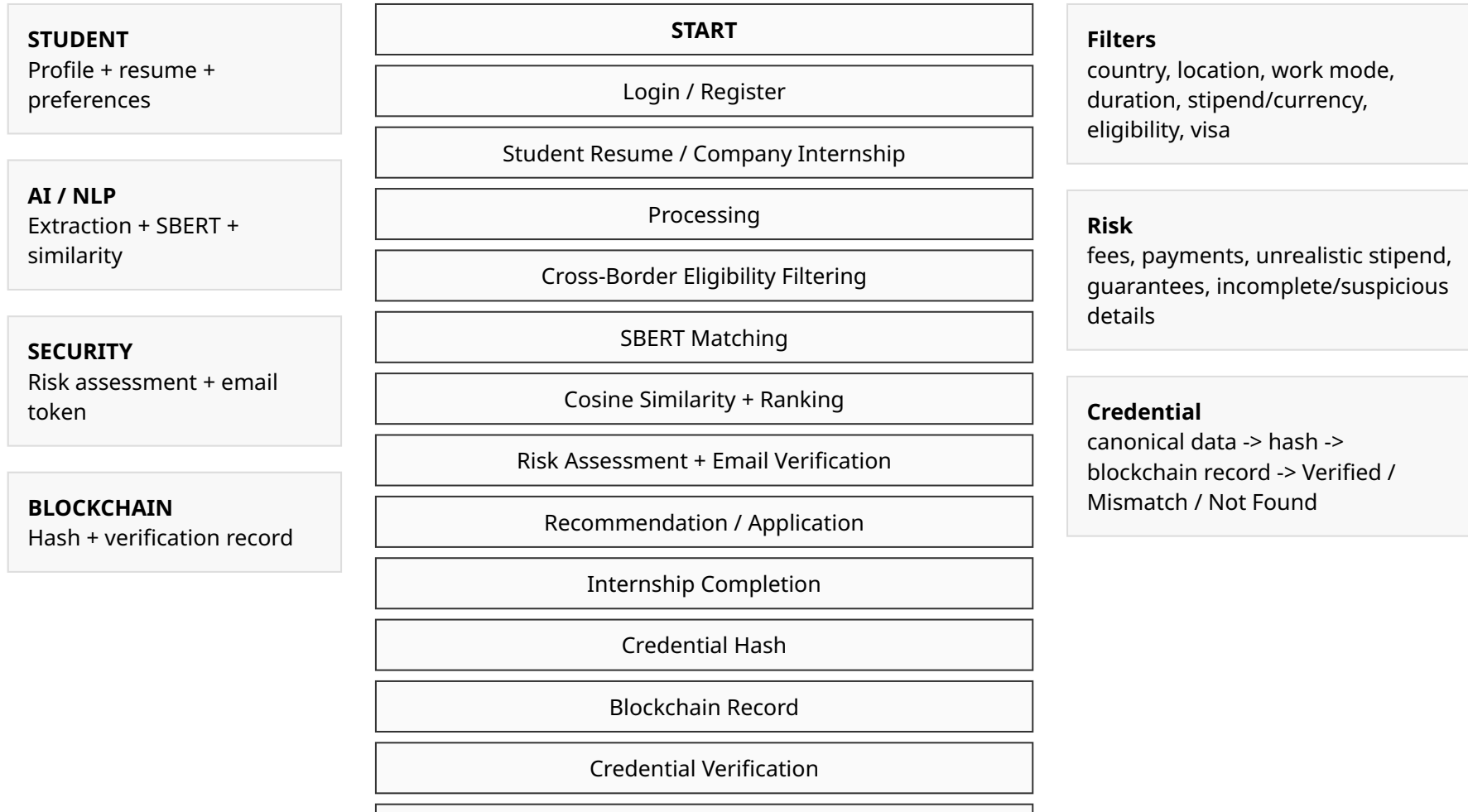
Objectives

- Develop an intelligent internship recommendation platform for students and internship providers.
- Utilize Sentence-BERT (SBERT) to understand the contextual meaning of resumes and internship descriptions.
- Apply Cosine Similarity to calculate compatibility scores and recommend relevant internships.
- Support cross-border internship filtering using location, duration, stipend, eligibility, and visa information.
- Implement suspicious internship detection and official company email verification.
- Integrate blockchain technology for secure and tamper-proof verification of internship credentials.

Proposed System

- Frontend: React.js + Tailwind CSS; Backend: Spring Boot REST API; Database: MySQL.
- Python AI service runs Sentence-BERT independently to generate resume and internship embeddings; cosine similarity provides a ranking signal.
- Student workflow: register/login -> profile and resume -> structured filters -> semantic matching -> ranked recommendations -> application.
- Company workflow: register -> official-email token verification -> create internship with international attributes -> risk assessment -> publish/flag -> manage applications.
- Credential workflow: completed internship credential -> canonicalize selected data -> generate cryptographic hash -> store identifier/hash and metadata on blockchain -> verify as Verified, Mismatch, or Not Found.
- Admin dashboard manages users, internships, flagged postings, verification records, and platform monitoring.

Flowchart



Algorithm

- Step 1: Student registers/logs in; company and admin roles are available for their respective workflows.
- Step 2: Student completes a profile and uploads a resume.
- Step 3: Resume text is extracted and cleaned/normalized for downstream NLP processing.
- Step 4: Skills, education, projects, experience, certifications, interests, and relevant constraints are represented in the student profile.
- Step 5: SBERT generates semantic embeddings for the student resume/profile and internship descriptions.

Algorithm

- Step 6: Cosine similarity is calculated between the resume/student representation and internship representations.
- Step 7: Eligible internships are ranked using the similarity score; the score is a ranking signal, not a guaranteed probability of suitability.
- Step 8: Suspicious-posting risk assessment calculates weighted indicators and returns Low, Medium, or High Risk with reasons.
- Step 9: Company official-email token is validated and the company is marked as having a verified email address.
- Step 10: Student views verified recommendations, opens details, and applies through the platform; application status is managed afterward.
- Step 11: After completion, the credential is canonicalized, hashed, recorded on blockchain, and later checked for Verified, Mismatch, or Not Found.

Conclusion

- The project delivers an integrated cross-border internship ecosystem spanning recommendation, verification, application, and credential workflows.
- SBERT and cosine similarity provide the semantic matching layer, while mandatory structured filters prevent practically unsuitable opportunities from being recommended.
- Transparent risk assessment surfaces potentially suspicious internship postings without treating a risk score as conclusive proof of fraud.
- Official company email verification establishes control of the submitted address before publishing; it is not an independent legal-identity check.
- Blockchain-backed credential records provide an integrity-verification path for completed internships.
- Final claims must be supported by actual implementation, testing evidence, evaluation metrics, and deployment results; invented results are not permitted.

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**Thank
You**